Experiment 3 Section 7/8 Replication: Trust and Interaction Experience Regression

Goal

Replicate Experiment 3 notebook Section 7 and Section 8 analysis structure with two regression outcomes:

- `composite_trust_score ~ vis_type * interaction * detail_presentation`
- `interaction_experience ~ vis_type * interaction * detail_presentation`

This report follows the same trust-survey preprocessing used in `analysis-0318.ipynb`, including:

- loading `0318`, `0319`, `0322` batches
- extracting trust/interaction items from `response`
- normalizing Likert items from `0-6` to `1-7` when needed
- excluding attention-check failures

Packages

```
required_packages <- c(
  "dplyr",
  "jsonlite",
  "knitr",
  "emmeans",
  "ggplot2",
  "lme4",
  "lmerTest"
)

missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]

if (length(missing_packages) > 0) {
  stop(
    "Missing required packages: ",
    paste(missing_packages, collapse = ", "),
    ". Install them before rendering."
  )
}

lapply(required_packages, library, character.only = TRUE)
```

```
## [[1]]
## [1] "dplyr"     "stats"     "graphics"  "grDevices" "utils"     "datasets"
## [7] "methods"   "base"
##
## [[2]]
## [1] "jsonlite"  "dplyr"     "stats"     "graphics"  "grDevices" "utils"
## [7] "datasets"  "methods"   "base"
##
## [[3]]
##  [1] "knitr"     "jsonlite"  "dplyr"     "stats"     "graphics"  "grDevices"
##  [7] "utils"     "datasets"  "methods"   "base"
##
## [[4]]
##  [1] "emmeans"   "knitr"     "jsonlite"  "dplyr"     "stats"     "graphics"
##  [7] "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[5]]
##  [1] "ggplot2"   "emmeans"   "knitr"     "jsonlite"  "dplyr"     "stats"
##  [7] "graphics"  "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[6]]
##  [1] "lme4"      "Matrix"    "ggplot2"   "emmeans"   "knitr"     "jsonlite"
##  [7] "dplyr"     "stats"     "graphics"  "grDevices" "utils"     "datasets"
## [13] "methods"   "base"
##
## [[7]]
##  [1] "lmerTest"  "lme4"      "Matrix"    "ggplot2"   "emmeans"   "knitr"
##  [7] "jsonlite"  "dplyr"     "stats"     "graphics"  "grDevices" "utils"
## [13] "datasets"  "methods"   "base"
```

Paths

```
report_dir <- local({
  current_input <- tryCatch(
    knitr::current_input(),
    error = function(e) character(0)
  )

  if (length(current_input) == 1 && nzchar(current_input)) {
    input_path <- current_input

    if (!grepl("^(/|[A-Za-z]:[/\\\\])", input_path)) {
      input_path <- file.path(getwd(), input_path)
    }

    if (grepl("\\.[Rr]md$", input_path) || file.exists(input_path)) {
      return(normalizePath(dirname(input_path), winslash = "/", mustWork = TRUE))
    }
  }

  fallback_dir <- file.path(getwd(), "Experiment3", "analysis")
  normalizePath(fallback_dir, winslash = "/", mustWork = TRUE)
})

resolve_report_path <- function(path_value) {
  if (grepl("^(/|[A-Za-z]:[/\\\\])", path_value)) {
    return(normalizePath(path_value, winslash = "/", mustWork = TRUE))
  }

  normalizePath(
    file.path(report_dir, path_value),
    winslash = "/",
    mustWork = TRUE
  )
}

data_dirs <- vapply(params$data_dirs, resolve_report_path, character(1))
```

Constants

```
BASELINE_CONDITION_NUMBER <- 1
TREATMENT_CONDITIONS <- c(21, 22, 23, 24, 25, 26)

INTERACTION_CONDITION_LABELS <- c(
  "1" = "Baseline",
  "21" = "Hover Show One",
  "22" = "Hover Show All",
  "23" = "Click Show One",
  "24" = "Click Show All",
  "25" = "Animation Show One",
  "26" = "Animation Show All"
)

SECTION7_INTERACTION_ORDER <- c(
  "Hover Show One",
  "Hover Show All",
  "Click Show One",
  "Click Show All",
  "Animation Show One",
  "Animation Show All"
)

TRUST_ITEM_KEYS <- c(
  "skeptical_rating",
  "data_trust",
  "usability_difficulty",
  "comprehension_ease"
)

INTERACTION_ITEM_KEYS <- c(
  "navigation_control",
  "content_control",
  "pace_control",
  "interface_exploration",
  "interface_responsiveness",
  "user_communication",
  "personal_conversation",
  "interface_interaction",
  "interface_sensitivity"
)
```

Helper Functions

```
extract_condition_number <- function(condition_id) {
  if (is.na(condition_id) || !nzchar(as.character(condition_id))) {
    return(NA_real_)
  }

  raw <- sub(".*condition_([0-9]+).*", "\\1", as.character(condition_id), perl = TRUE)
  if (!grepl("^[0-9]+$", raw)) {
    return(NA_real_)
  }

  suppressWarnings(as.numeric(raw))
}

map_interaction_group <- function(condition_number) {
  key <- as.character(suppressWarnings(as.integer(condition_number)))
  if (is.na(condition_number) || !(key %in% names(INTERACTION_CONDITION_LABELS))) {
    return("Other")
  }
  INTERACTION_CONDITION_LABELS[[key]]
}

map_visualization_group <- function(condition_number, visualization_technique, condition_id, condition_name) {
  cn <- suppressWarnings(as.integer(condition_number))
  if (!is.na(cn) && cn == BASELINE_CONDITION_NUMBER) {
    return("Baseline")
  }

  technique_raw <- tolower(trimws(ifelse(is.na(visualization_technique), "", visualization_technique)))
  if (technique_raw == "baseline") {
    return("Baseline")
  }
  if (technique_raw == "confidence_interval") {
    return("PI")
  }
  if (technique_raw == "ensemble_plot") {
    return("Ensemble")
  }
  if (technique_raw == "combined_plot") {
    return("PI + Ensemble")
  }

  condition_id_raw <- tolower(trimws(ifelse(is.na(condition_id), "", condition_id)))
  if (grepl("_ci$", condition_id_raw)) {
    return("PI")
  }
  if (grepl("_ep$", condition_id_raw)) {
    return("Ensemble")
  }
  if (grepl("_ciep$", condition_id_raw)) {
    return("PI + Ensemble")
  }

  condition_name_raw <- tolower(trimws(ifelse(is.na(condition_name), "", condition_name)))
  if (grepl("baseline", condition_name_raw)) {
    return("Baseline")
  }
  if (grepl("combined", condition_name_raw)) {
    return("PI + Ensemble")
  }
  if (grepl("ensemble", condition_name_raw)) {
    return("Ensemble")
  }
  if (grepl("\\(ci\\)|confidence", condition_name_raw)) {
    return("PI")
  }

  "Other"
}

parse_response_json <- function(value) {
  if (is.na(value) || !nzchar(as.character(value))) {
    return(list())
  }

  parsed <- tryCatch(
    jsonlite::fromJSON(as.character(value), simplifyVector = TRUE),
```

```
    error = function(e) list()
  )

  if (is.null(parsed) || !is.list(parsed)) {
    return(list())
  }

  parsed
}

expand_response_columns <- function(data, response_col = "response") {
  if (!(response_col %in% names(data))) {
    return(data)
  }

  parsed <- lapply(data[[response_col]], parse_response_json)
  all_keys <- unique(unlist(lapply(parsed, names), use.names = FALSE))
  all_keys <- all_keys[!is.na(all_keys) & nzchar(all_keys)]

  out <- data

  for (key in all_keys) {
    extracted <- vapply(
      parsed,
      function(item) {
        value <- item[[key]]
        if (is.null(value) || length(value) == 0) {
          return(NA_character_)
        }
        as.character(value[[1]])
      },
      character(1)
    )

    if (key %in% names(out)) {
      existing <- out[[key]]
      fill_idx <- is.na(existing) | trimws(as.character(existing)) == ""
      existing[fill_idx] <- extracted[fill_idx]
      out[[key]] <- existing
    } else {
      out[[key]] <- extracted
    }
  }

  out
}

normalize_likert_0_6_to_1_7 <- function(values) {
  numeric_values <- suppressWarnings(as.numeric(values))
  valid <- numeric_values[!is.na(numeric_values)]

  if (length(valid) == 0) {
    return(numeric_values)
  }

  if (all(valid >= 0 & valid <= 6)) {
    return(pmin(pmax(numeric_values + 1, 1), 7))
  }

  numeric_values
}

set_sum_contrasts <- function(data, factors) {
  out <- data
  for (factor_name in factors) {
    out[[factor_name]] <- factor(out[[factor_name]])
    contrasts(out[[factor_name]]) <- contr.sum(length(levels(out[[factor_name]])))
  }
  out
}

identify_attention_check_failures <- function(data) {
  if (!all(c("trial_type", "participant_id") %in% names(data))) {
    return(character(0))
  }

  survey_rows <- data %>%
    dplyr::filter(trial_type == "trust-survey") %>%
    expand_response_columns("response")
```

```
  if (nrow(survey_rows) == 0) {
    return(character(0))
  }

  for (key in c(TRUST_ITEM_KEYS, INTERACTION_ITEM_KEYS)) {
    if (!(key %in% names(survey_rows))) {
      survey_rows[[key]] <- NA_real_
    }
    survey_rows[[key]] <- normalize_likert_0_6_to_1_7(survey_rows[[key]])
  }

  trust_complete <- rowSums(!is.na(survey_rows[, TRUST_ITEM_KEYS, drop = FALSE])) == length(TRUST_ITEM_KEYS)
  interaction_complete <- rowSums(!is.na(survey_rows[, INTERACTION_ITEM_KEYS, drop = FALSE])) == length(INTERACTI
ON_ITEM_KEYS)

  trust_uniform <- apply(survey_rows[, TRUST_ITEM_KEYS, drop = FALSE], 1, function(x) {
    length(unique(stats::na.omit(x))) == 1
  })

  interaction_uniform <- apply(survey_rows[, INTERACTION_ITEM_KEYS, drop = FALSE], 1, function(x) {
    length(unique(stats::na.omit(x))) == 1
  })

  failed <- survey_rows$participant_id[trust_complete & interaction_complete & trust_uniform & interaction_unifor
m]
  unique(as.character(failed[!is.na(failed)]))
}

map_interaction_factor <- function(condition_number) {
  dplyr::case_when(
    condition_number %in% c(21, 22) ~ "hover",
    condition_number %in% c(23, 24) ~ "click",
    condition_number %in% c(25, 26) ~ "animation",
    TRUE ~ NA_character_
  )
}

map_detail_factor <- function(condition_number) {
  dplyr::case_when(
    condition_number %in% c(21, 23, 25) ~ "show_one",
    condition_number %in% c(22, 24, 26) ~ "show_all",
    TRUE ~ NA_character_
  )
}

map_vis_type <- function(visualization_group) {
  dplyr::case_when(
    visualization_group == "PI" ~ "CI",
    visualization_group == "Ensemble" ~ "EP",
    visualization_group == "PI + Ensemble" ~ "CIEP",
    TRUE ~ NA_character_
  )
}

read_participant_csv <- function(file_path) {
  df <- read.csv(file_path, stringsAsFactors = FALSE, check.names = FALSE)
  df$source_file <- basename(file_path)

  if (!("participant_id" %in% names(df)) || all(is.na(df$participant_id))) {
    token <- strsplit(tools::file_path_sans_ext(basename(file_path)), "_")[[1]]
    inferred_id <- ifelse(length(token) >= 2, token[[2]], NA_character_)
    df$participant_id <- inferred_id
  }

  if (!("condition_id" %in% names(df))) {
    df$condition_id <- NA_character_
  }
  if (!("condition_name" %in% names(df))) {
    df$condition_name <- NA_character_
  }

  df$condition_number <- vapply(df$condition_id, extract_condition_number, numeric(1))
  df$interaction_group <- vapply(df$condition_number, map_interaction_group, character(1))
  df$visualization_group <- mapply(
    map_visualization_group,
    condition_number = df$condition_number,
    visualization_technique = if ("visualization_technique" %in% names(df)) df$visualization_technique else NA_ch
aracter_,
```

```
    condition_id = df$condition_id,
    condition_name = df$condition_name,
    USE.NAMES = FALSE
  )

  df
}
```

Load and Filter Data

```
csv_files <- unlist(
  lapply(
    data_dirs,
    function(dir_path) {
      list.files(dir_path, pattern = params$csv_pattern, full.names = TRUE)
    }
  ),
  use.names = FALSE
)

if (length(csv_files) == 0) {
  stop(
    "No CSV files found in: ",
    paste(data_dirs, collapse = ", "),
    " with pattern ",
    params$csv_pattern
  )
}

combined_data <- dplyr::bind_rows(lapply(csv_files, read_participant_csv))

before_n <- dplyr::n_distinct(combined_data$participant_id)

combined_data <- combined_data %>%
  dplyr::filter(!is.na(participant_id)) %>%
  dplyr::filter(tolower(as.character(participant_id)) != "test")

attention_failures <- identify_attention_check_failures(combined_data)

if (length(attention_failures) > 0) {
  combined_data <- combined_data %>%
    dplyr::filter(!(as.character(participant_id) %in% attention_failures))
}

after_n <- dplyr::n_distinct(combined_data$participant_id)

cat("Data directories:\n")
```

```
## Data directories:
```

```
cat(paste0("  - ", data_dirs), sep = "\n")
```

```
##   - /Users/songwen/Documents/Code Lab/Interact4Trust/Experiment3/data/0318/data
##   - /Users/songwen/Documents/Code Lab/Interact4Trust/Experiment3/data/0319/data
##   - /Users/songwen/Documents/Code Lab/Interact4Trust/Experiment3/data/0322/data
```

```
cat("\nCSV files loaded: ", length(csv_files), "\n", sep = "")
```

```
##
## CSV files loaded: 142
```

```
cat("Participants: ", before_n, " -> ", after_n, "\n", sep = "")
```

```
## Participants: 142 -> 140
```

```
cat("Attention-check exclusions: ", length(attention_failures), "\n", sep = "")
```

```
## Attention-check exclusions: 2
```

Trust-Survey Preprocessing

```
trust_survey <- combined_data %>%
  dplyr::filter(trial_type == "trust-survey") %>%
  expand_response_columns("response")

for (key in c(TRUST_ITEM_KEYS, INTERACTION_ITEM_KEYS)) {
  if (!(key %in% names(trust_survey))) {
    trust_survey[[key]] <- NA_real_
  }
  trust_survey[[key]] <- normalize_likert_0_6_to_1_7(trust_survey[[key]])
}

trust_survey <- trust_survey %>%
  dplyr::mutate(
    skeptical_rating_rev = 8 - as.numeric(skeptical_rating),
    usability_difficulty_rev = 8 - as.numeric(usability_difficulty),
    composite_trust_score = rowMeans(
      cbind(
        skeptical_rating_rev,
        as.numeric(data_trust),
        usability_difficulty_rev,
        as.numeric(comprehension_ease)
      ),
      na.rm = TRUE
    ),
    interaction_experience = rowMeans(
      dplyr::across(dplyr::all_of(INTERACTION_ITEM_KEYS)),
      na.rm = TRUE
    ),
    interaction = map_interaction_factor(condition_number),
    detail_presentation = map_detail_factor(condition_number),
    vis_type = map_vis_type(visualization_group),
    interaction_group = vapply(condition_number, map_interaction_group, character(1))
  )

trust_survey$composite_trust_score[is.nan(trust_survey$composite_trust_score)] <- NA_real_
trust_survey$interaction_experience[is.nan(trust_survey$interaction_experience)] <- NA_real_

model_input_raw <- trust_survey %>%
  dplyr::filter(condition_number %in% TREATMENT_CONDITIONS) %>%
  dplyr::filter(
    !is.na(interaction),
    !is.na(detail_presentation),
    !is.na(vis_type),
    !is.na(composite_trust_score),
    !is.na(interaction_experience)
  )

# Aggregate if duplicate rows exist for a participant x condition cell.
model_input <- model_input_raw %>%
  dplyr::group_by(participant_id, vis_type, interaction, detail_presentation) %>%
  dplyr::summarise(
    composite_trust_score = mean(composite_trust_score, na.rm = TRUE),
    interaction_experience = mean(interaction_experience, na.rm = TRUE),
    .groups = "drop"
  )

# Complete within-subject grid for interaction (3) x detail_presentation (2) = 6 cells.
participant_cell_counts <- model_input %>%
  dplyr::count(participant_id, name = "n_cells")

complete_participants <- participant_cell_counts %>%
  dplyr::filter(n_cells == 6) %>%
  dplyr::pull(participant_id)

model_data <- model_input %>%
  dplyr::filter(participant_id %in% complete_participants) %>%
  dplyr::mutate(
    participant_id = factor(participant_id),
    interaction = factor(interaction, levels = c("hover", "click", "animation")),
    detail_presentation = factor(detail_presentation, levels = c("show_one", "show_all")),
    vis_type = factor(vis_type, levels = c("CI", "EP", "CIEP"))
  )

cat("Trust-survey rows (raw): ", nrow(trust_survey), "\n", sep = "")
```

```
## Trust-survey rows (raw): 980
```

```
cat("Model rows after filtering: ", nrow(model_data), "\n", sep = "")
```

```
## Model rows after filtering: 840
```

```
cat("Model participants (complete 6-cell within-subject grid): ", dplyr::n_distinct(model_data$participant_id),
"\n", sep = "")
```

```
## Model participants (complete 6-cell within-subject grid): 140
```

Section 7: Cross-Interaction Technique Summary

```
section7_summary <- model_data %>%
  dplyr::group_by(interaction, detail_presentation) %>%
  dplyr::summarise(
    n = dplyr::n(),
    trust_mean = mean(composite_trust_score, na.rm = TRUE),
    trust_sd = stats::sd(composite_trust_score, na.rm = TRUE),
    trust_se = trust_sd / sqrt(n),
    interaction_mean = mean(interaction_experience, na.rm = TRUE),
    interaction_sd = stats::sd(interaction_experience, na.rm = TRUE),
    interaction_se = interaction_sd / sqrt(n),
    .groups = "drop"
  ) %>%
  dplyr::arrange(interaction, detail_presentation)

knitr::kable(section7_summary, digits = 3, caption = "Section 7 summary by interaction x detail presentation")
```

Section 7 summary by interaction x detail presentation

interaction	detail_presentation	n	trust_mean	trust_sd	trust_se	interaction_mean	interaction_sd	interacti
hover	show_one	140	4.870	1.087	0.092	3.928	1.087	
hover	show_all	140	4.825	1.155	0.098	3.805	1.099	
click	show_one	140	5.002	1.152	0.097	4.097	1.104	
click	show_all	140	4.850	1.185	0.100	3.863	1.032	
animation	show_one	140	4.450	1.297	0.110	3.201	1.288	
animation	show_all	140	4.259	1.300	0.110	3.009	1.225	

Section 8: Interaction x Visualization Grid Summary

```
section8_summary <- model_data %>%
  dplyr::group_by(vis_type, interaction, detail_presentation) %>%
  dplyr::summarise(
    n = dplyr::n(),
    trust_mean = mean(composite_trust_score, na.rm = TRUE),
    trust_sd = stats::sd(composite_trust_score, na.rm = TRUE),
    trust_se = trust_sd / sqrt(n),
    interaction_mean = mean(interaction_experience, na.rm = TRUE),
    interaction_sd = stats::sd(interaction_experience, na.rm = TRUE),
    interaction_se = interaction_sd / sqrt(n),
    .groups = "drop"
  ) %>%
  dplyr::arrange(vis_type, interaction, detail_presentation)

knitr::kable(section8_summary, digits = 3, caption = "Section 8 summary by vis_type x interaction condition")
```

Section 8 summary by vis_type x interaction condition

vis_type	interaction	detail_presentation	n	trust_mean	trust_sd	trust_se	interaction_mean	interaction_sd
CI	hover	show_one	49	4.847	1.053	0.150	3.964	1.102
CI	hover	show_all	49	4.842	1.262	0.180	3.791	1.070
CI	click	show_one	49	5.128	1.189	0.170	4.193	1.055
CI	click	show_all	49	5.061	1.102	0.157	3.905	0.984
CI	animation	show_one	49	4.556	1.498	0.214	3.240	1.393
CI	animation	show_all	49	4.434	1.268	0.181	3.091	1.235
EP	hover	show_one	45	5.028	0.972	0.145	3.891	1.031
EP	hover	show_all	45	4.883	1.131	0.169	3.751	1.097
EP	click	show_one	45	4.944	1.020	0.152	4.015	0.987
EP	click	show_all	45	4.761	1.079	0.161	3.726	1.065
EP	animation	show_one	45	4.439	1.150	0.171	3.165	1.282

vis_type	interaction	detail_presentation	n	trust_mean	trust_sd	trust_se	interaction_mean	interaction_sd
EP	animation	show_all	45	4.483	1.219	0.182	3.079	1.300
CIEP	hover	show_one	46	4.739	1.226	0.181	3.925	1.145
CIEP	hover	show_all	46	4.750	1.079	0.159	3.872	1.152
CIEP	click	show_one	46	4.924	1.243	0.183	4.075	1.267
CIEP	click	show_all	46	4.712	1.351	0.199	3.954	1.057
CIEP	animation	show_one	46	4.348	1.220	0.180	3.193	1.202
CIEP	animation	show_all	46	3.853	1.340	0.198	2.853	1.147

Regression 1: Trust

```
regression_model_data <- set_sum_contrasts(
  model_data,
  factors = c("vis_type", "interaction", "detail_presentation")
)

trust_lm <- lm(
  composite_trust_score ~ vis_type + interaction + detail_presentation,
  data = regression_model_data,
  na.action = na.exclude
)

trust_anova <- anova(trust_lm)
trust_anova
```

```
## Analysis of Variance Table
##
## Response: composite_trust_score
##                      Df  Sum Sq Mean Sq F value    Pr(>F)
## vis_type              2   10.28  5.1410  3.6075   0.02754 *
## interaction           2   53.72 26.8619 18.8492 9.852e-09 ***
## detail_presentation   1    3.50  3.5036  2.4585   0.11727
## Residuals           834 1188.53  1.4251
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Regression 1: Trust (Joint Effects)

```
trust_joint_lm <- lm(
  composite_trust_score ~ vis_type * interaction * detail_presentation,
  data = regression_model_data,
  na.action = na.exclude
)

trust_joint_anova <- anova(trust_joint_lm)
trust_joint_anova
```

```
## Analysis of Variance Table
##
## Response: composite_trust_score
##                                          Df  Sum Sq Mean Sq F value    Pr(>F)
## vis_type                                  2   10.28  5.1410  3.5854   0.02816
## interaction                               2   53.72 26.8619 18.7336 1.107e-08
## detail_presentation                       1    3.50  3.5036  2.4435   0.11840
## vis_type:interaction                      4    4.97  1.2421  0.8662   0.48371
## vis_type:detail_presentation              2    1.12  0.5578  0.3890   0.67786
## interaction:detail_presentation           2    0.80  0.4021  0.2804   0.75554
## vis_type:interaction:detail_presentation  4    2.98  0.7461  0.5203   0.72085
## Residuals                               822 1178.66  1.4339
##
## vis_type                                *
## interaction                             ***
## detail_presentation
## vis_type:interaction
## vis_type:detail_presentation
## interaction:detail_presentation
## vis_type:interaction:detail_presentation
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Regression 2: Interaction Experience

```
interaction_lm <- lm(
  interaction_experience ~ vis_type + interaction + detail_presentation,
  data = regression_model_data,
  na.action = na.exclude
)

interaction_anova <- anova(interaction_lm)
interaction_anova
```

```
## Analysis of Variance Table
##
## Response: interaction_experience
##                      Df  Sum Sq Mean Sq F value  Pr(>F)
## vis_type              2    1.22   0.611  0.4682 0.62631
## interaction           2  126.86  63.429 48.6262 < 2e-16 ***
## detail_presentation   1    7.02   7.018  5.3799 0.02061 *
## Residuals           834 1087.88   1.304
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Regression 2: Interaction Experience (Joint Effects)

```
interaction_joint_lm <- lm(
  interaction_experience ~ vis_type * interaction * detail_presentation,
  data = regression_model_data,
  na.action = na.exclude
)

interaction_joint_anova <- anova(interaction_joint_lm)
interaction_joint_anova
```

```
## Analysis of Variance Table
##
## Response: interaction_experience
##                                          Df  Sum Sq Mean Sq F value  Pr(>F)
## vis_type                                  2    1.22   0.611  0.4630 0.62958
## interaction                               2  126.86  63.429 48.0849 < 2e-16
## detail_presentation                       1    7.02   7.018  5.3200 0.02133
## vis_type:interaction                      4    1.73   0.433  0.3281 0.85919
## vis_type:detail_presentation              2    0.05   0.024  0.0181 0.98211
## interaction:detail_presentation           2    0.43   0.217  0.1649 0.84803
## vis_type:interaction:detail_presentation  4    1.37   0.342  0.2592 0.90405
## Residuals                               822 1084.30   1.319
##
## vis_type
## interaction                             ***
## detail_presentation                     *
## vis_type:interaction
## vis_type:detail_presentation
## interaction:detail_presentation
## vis_type:interaction:detail_presentation
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

EMMs: Interaction

```
extract_emmean_with_sd <- function(model, factor_name, sd_data, value_col) {
  emmean_tbl <- emmeans::emmeans(
    model,
    specs = stats::as.formula(paste0("~ ", factor_name))
  ) %>%
    as.data.frame() %>%
    dplyr::select(dplyr::all_of(factor_name), emmean, SE)

  sd_tbl <- sd_data %>%
    dplyr::group_by(.data[[factor_name]]) %>%
    dplyr::summarise(
      sd = stats::sd(.data[[value_col]], na.rm = TRUE),
      .groups = "drop"
    )

  emmean_tbl %>%
    dplyr::left_join(sd_tbl, by = factor_name)
}

interaction_level_scores <- model_data %>%
  dplyr::group_by(participant_id, interaction) %>%
  dplyr::summarise(
    composite_trust_score = mean(composite_trust_score, na.rm = TRUE),
    interaction_experience = mean(interaction_experience, na.rm = TRUE),
    .groups = "drop"
  )

trust_interaction_emmeans <- extract_emmean_with_sd(
  model = trust_joint_lm,
  factor_name = "interaction",
  sd_data = interaction_level_scores,
  value_col = "composite_trust_score"
)

interactionexp_interaction_emmeans <- extract_emmean_with_sd(
  model = interaction_joint_lm,
  factor_name = "interaction",
  sd_data = interaction_level_scores,
  value_col = "interaction_experience"
)

knitr::kable(trust_interaction_emmeans, digits = 4, caption = "EMMs (emmean, SE, SD) for interaction (trust)")
```

EMMs (emmean, SE, SD) for interaction (trust)

interaction	emmean	SE	sd
hover	4.8482	0.0716	0.9962
click	4.9217	0.0716	1.0883
animation	4.3522	0.0716	1.2025

```
knitr::kable(interactionexp_interaction_emmeans, digits = 4, caption = "EMMs (emmean, SE, SD) for interaction (in
teraction experience)")
```

EMMs (emmean, SE, SD) for interaction (interaction experience)

interaction	emmean	SE	sd
hover	3.8657	0.0687	1.0179
click	3.9779	0.0687	1.0235
animation	3.1036	0.0687	1.2054

EMMs: Detail Presentation

```
detail_level_scores <- model_data %>%
  dplyr::group_by(participant_id, detail_presentation) %>%
  dplyr::summarise(
    composite_trust_score = mean(composite_trust_score, na.rm = TRUE),
    interaction_experience = mean(interaction_experience, na.rm = TRUE),
    .groups = "drop"
  )

trust_detail_emmeans <- extract_emmean_with_sd(
  model = trust_joint_lm,
  factor_name = "detail_presentation",
  sd_data = detail_level_scores,
  value_col = "composite_trust_score"
)

interactionexp_detail_emmeans <- extract_emmean_with_sd(
  model = interaction_joint_lm,
  factor_name = "detail_presentation",
  sd_data = detail_level_scores,
  value_col = "interaction_experience"
)

knitr::kable(trust_detail_emmeans, digits = 4, caption = "EMMs (emmean, SE, SD) for detail_presentation (trust)")
```

EMMs (emmean, SE, SD) for detail_presentation (trust)

detail_presentation	emmean	SE	sd
show_one	4.7725	0.0585	0.9944
show_all	4.6422	0.0585	1.0237

```
knitr::kable(interactionexp_detail_emmeans, digits = 4, caption = "EMMs (emmean, SE, SD) for detail_presentation
(interaction experience)")
```

EMMs (emmean, SE, SD) for detail_presentation (interaction experience)

detail_presentation	emmean	SE	sd
show_one	3.7402	0.0561	1.0216
show_all	3.5579	0.0561	0.9853

Additional Analysis: Q1-Q3 (Baseline + Click/Hover x Detail

Presentation x Visualization Type)

```
prediction_q123_raw <- combined_data %>%
  dplyr::filter(trial_type == "prediction-task") %>%
  dplyr::mutate(
    is_baseline = condition_number == BASELINE_CONDITION_NUMBER,
    probability = suppressWarnings(as.numeric(probability_estimate)),
    city_a_estimate = suppressWarnings(as.numeric(city_a_estimate)),
    city_b_estimate = suppressWarnings(as.numeric(city_b_estimate)),
    estimated_value_row = rowMeans(cbind(city_a_estimate, city_b_estimate), na.rm = TRUE),
    estimated_value_row = ifelse(is.finite(estimated_value_row), estimated_value_row, NA_real_),
    interaction_mode = dplyr::case_when(
      is_baseline ~ "hover",
      TRUE ~ map_interaction_factor(condition_number)
    ),
    detail_presentation = dplyr::case_when(
      is_baseline ~ "show_one",
      TRUE ~ map_detail_factor(condition_number)
    ),
    vistype = dplyr::case_when(
      is_baseline ~ "CI",
      TRUE ~ map_vis_type(visualization_group)
    ),
    decision_result_row = dplyr::case_when(
      !is.na(travel_choice) & !is.na(city_a_label) & travel_choice == city_a_label ~ "A",
      !is.na(travel_choice) & !is.na(city_b_label) & travel_choice == city_b_label ~ "B",
      tolower(trimws(as.character(travel_choice))) == "no preference" ~ "No Preference",
      TRUE ~ NA_character_
    )
  ) %>%
  dplyr::filter(condition_number %in% c(BASELINE_CONDITION_NUMBER, TREATMENT_CONDITIONS)) %>%
  dplyr::filter(interaction_mode %in% c("hover", "click")) %>%
  dplyr::filter(!is.na(vistype), !is.na(detail_presentation), !is.na(interaction_mode))

mode_or_na <- function(x) {
  x <- as.character(x)
  x <- x[!is.na(x) & nzchar(x)]
  if (length(x) == 0) {
    return(NA_character_)
  }
  names(sort(table(x), decreasing = TRUE))[1]
}

prediction_q123_condition <- prediction_q123_raw %>%
  dplyr::group_by(participant_id, condition_number, condition_name, vistype, interaction_mode, detail_presentatio
n) %>%
  dplyr::summarise(
    probability = mean(probability, na.rm = TRUE),
    estimated_value = mean(estimated_value_row, na.rm = TRUE),
    decision_result = mode_or_na(decision_result_row),
    .groups = "drop"
  ) %>%
  dplyr::mutate(
    probability = ifelse(is.finite(probability), probability, NA_real_),
    estimated_value = ifelse(is.finite(estimated_value), estimated_value, NA_real_),
    participant_id = factor(participant_id),
    is_baseline = factor(ifelse(condition_number == BASELINE_CONDITION_NUMBER, "Yes", "No"), levels = c("No", "Ye
s")),
    vistype = factor(vistype, levels = c("CI", "EP", "CIEP")),
    interaction_mode = factor(interaction_mode, levels = c("hover", "click")),
    detail_presentation = factor(detail_presentation, levels = c("show_one", "show_all")),
    decision_result = factor(decision_result, levels = c("A", "B", "No Preference"))
  ) %>%
  droplevels()

# Require complete within-subject grid for baseline + click/hover x detail (1 + 2 x 2 = 5 cells).
q123_participant_cell_counts <- prediction_q123_condition %>%
  dplyr::count(participant_id, name = "n_cells")

q123_complete_participants <- q123_participant_cell_counts %>%
  dplyr::filter(n_cells == 5) %>%
  dplyr::pull(participant_id)

prediction_q123_model_base <- prediction_q123_condition %>%
  dplyr::filter(participant_id %in% q123_complete_participants) %>%
  droplevels()

q123_q1_data <- prediction_q123_model_base %>%
```

```
  dplyr::filter(is.finite(probability))

q123_q2_data <- prediction_q123_model_base %>%
  dplyr::filter(is.finite(estimated_value))

q123_q3_data <- prediction_q123_model_base %>%
  dplyr::filter(!is.na(decision_result)) %>%
  droplevels()

cat("Q1/Q2/Q3 rows after filtering: ", nrow(prediction_q123_model_base), "\n", sep = "")
```

```
## Q1/Q2/Q3 rows after filtering: 700
```

```
cat("Q1/Q2/Q3 participants (complete 5-cell grid): ", dplyr::n_distinct(prediction_q123_model_base$participant_i
d), "\n", sep = "")
```

```
## Q1/Q2/Q3 participants (complete 5-cell grid): 140
```

Q1: Probability (Continuous)

```
q123_q1_model_data <- set_sum_contrasts(
  q123_q1_data,
  factors = c("vistype", "interaction_mode", "detail_presentation", "is_baseline")
)

# Repeated-measures best-practice model
q123_q1_lmer <- lmer(
  probability ~ vistype * interaction_mode * detail_presentation * is_baseline + (1 | participant_id),
  data = q123_q1_model_data,
  REML = FALSE,
  na.action = na.exclude
)

anova(q123_q1_lmer, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                                                          Sum Sq Mean Sq NumDF
## vistype                                                  320.65  160.33     2
## interaction_mode                                          41.87   41.87     1
## detail_presentation                                       22.84   22.84     1
## is_baseline                                             1451.41 1451.41     1
## vistype:interaction_mode                                  20.65   10.33     2
## vistype:detail_presentation                              658.58  329.29     2
## interaction_mode:detail_presentation                     629.50  629.50     1
## vistype:interaction_mode:detail_presentation             509.06  254.53     2
## vistype:is_baseline
## interaction_mode:is_baseline
## detail_presentation:is_baseline
## vistype:interaction_mode:is_baseline
## vistype:detail_presentation:is_baseline
## interaction_mode:detail_presentation:is_baseline
## vistype:interaction_mode:detail_presentation:is_baseline
##                                                          DenDF F value  Pr(>F)
## vistype                                                 340.22  0.4753 0.62214
## interaction_mode                                        559.41  0.1241 0.72475
## detail_presentation                                     559.41  0.0677 0.79482
## is_baseline                                             678.20  4.3024 0.03844
## vistype:interaction_mode                                559.41  0.0306 0.96986
## vistype:detail_presentation                             559.41  0.9761 0.37742
## interaction_mode:detail_presentation                    559.41  1.8660 0.17248
## vistype:interaction_mode:detail_presentation            559.41  0.7545 0.47073
## vistype:is_baseline
## interaction_mode:is_baseline
## detail_presentation:is_baseline
## vistype:interaction_mode:is_baseline
## vistype:detail_presentation:is_baseline
## interaction_mode:detail_presentation:is_baseline
## vistype:interaction_mode:detail_presentation:is_baseline
##
## vistype
## interaction_mode
## detail_presentation
## is_baseline                                             *
## vistype:interaction_mode
## vistype:detail_presentation
## interaction_mode:detail_presentation
## vistype:interaction_mode:detail_presentation
## vistype:is_baseline
## interaction_mode:is_baseline
## detail_presentation:is_baseline
## vistype:interaction_mode:is_baseline
## vistype:detail_presentation:is_baseline
## interaction_mode:detail_presentation:is_baseline
## vistype:interaction_mode:detail_presentation:is_baseline
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
q123_q1_baseline_emm <- as.data.frame(
  summary(
    emmeans(
      q123_q1_lmer,
      ~ 1,
      at = list(
        is_baseline = "Yes",
        interaction_mode = "hover",
        detail_presentation = "show_one",
        vistype = "CI"
      )
    ),
    infer = c(TRUE, FALSE)
  )
) %>%
  dplyr::mutate(
    condition_label = "Baseline",
    is_baseline = "Yes"
  )

q123_q1_nonbaseline_emm <- as.data.frame(
  summary(
    emmeans(
      q123_q1_lmer,
      ~ 1,
      at = list(is_baseline = "No"),
      weights = "proportional"
    ),
    infer = c(TRUE, FALSE)
  )
) %>%
  dplyr::mutate(
    condition_label = "Non-baseline",
    is_baseline = "No"
  )

q123_q1_condition_emmeans <- dplyr::bind_rows(
  q123_q1_baseline_emm,
  q123_q1_nonbaseline_emm
) %>%
  dplyr::mutate(
    condition_label = factor(condition_label, levels = c("Baseline", "Non-baseline"))
  ) %>%
  dplyr::arrange(condition_label) %>%
  dplyr::select(condition_label, is_baseline, emmean, SE, df, lower.CL, upper.CL)

knitr::kable(
  q123_q1_condition_emmeans,
  digits = 3,
  caption = "Q1 EMMs for Baseline vs Non-baseline (Non-baseline averaged across conditions)"
)
```

Q1 EMMs for Baseline vs Non-baseline (Non-baseline averaged across conditions)

condition_label	is_baseline	emmean	SE	df	lower.CL	upper.CL
Baseline	Yes	52.429	1.793	580.926	48.906	55.951
Non-baseline	No	48.223	1.172	170.490	45.910	50.537

Q2: Estimated Prediction Value (Continuous)

```
q123_q2_model_data <- set_sum_contrasts(
  q123_q2_data,
  factors = c("vistype", "interaction_mode", "detail_presentation", "is_baseline")
)

# Repeated-measures best-practice model
q123_q2_lmer <- lmer(
  estimated_value ~ vistype * interaction_mode * detail_presentation * is_baseline + (1 | participant_id),
  data = q123_q2_model_data,
  REML = FALSE,
  na.action = na.exclude
)

anova(q123_q2_lmer, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                                                          Sum Sq Mean Sq NumDF
## vistype                                                 123.222  61.611     2
## interaction_mode                                          8.071   8.071     1
## detail_presentation                                     105.656 105.656     1
## is_baseline                                               5.950   5.950     1
## vistype:interaction_mode                                  8.959   4.479     2
## vistype:detail_presentation                               0.282   0.141     2
## interaction_mode:detail_presentation                      9.787   9.787     1
## vistype:interaction_mode:detail_presentation             66.159  33.079     2
## vistype:is_baseline
## interaction_mode:is_baseline
## detail_presentation:is_baseline
## vistype:interaction_mode:is_baseline
## vistype:detail_presentation:is_baseline
## interaction_mode:detail_presentation:is_baseline
## vistype:interaction_mode:detail_presentation:is_baseline
##                                                          DenDF F value Pr(>F)
## vistype                                                 214.57  0.3568 0.7003
## interaction_mode                                        557.62  0.0467 0.8289
## detail_presentation                                     557.62  0.6119 0.4344
## is_baseline                                             656.16  0.0345 0.8528
## vistype:interaction_mode                                557.62  0.0259 0.9744
## vistype:detail_presentation                             557.62  0.0008 0.9992
## interaction_mode:detail_presentation                    557.62  0.0567 0.8119
## vistype:interaction_mode:detail_presentation            557.62  0.1916 0.8257
## vistype:is_baseline
## interaction_mode:is_baseline
## detail_presentation:is_baseline
## vistype:interaction_mode:is_baseline
## vistype:detail_presentation:is_baseline
## interaction_mode:detail_presentation:is_baseline
## vistype:interaction_mode:detail_presentation:is_baseline
```

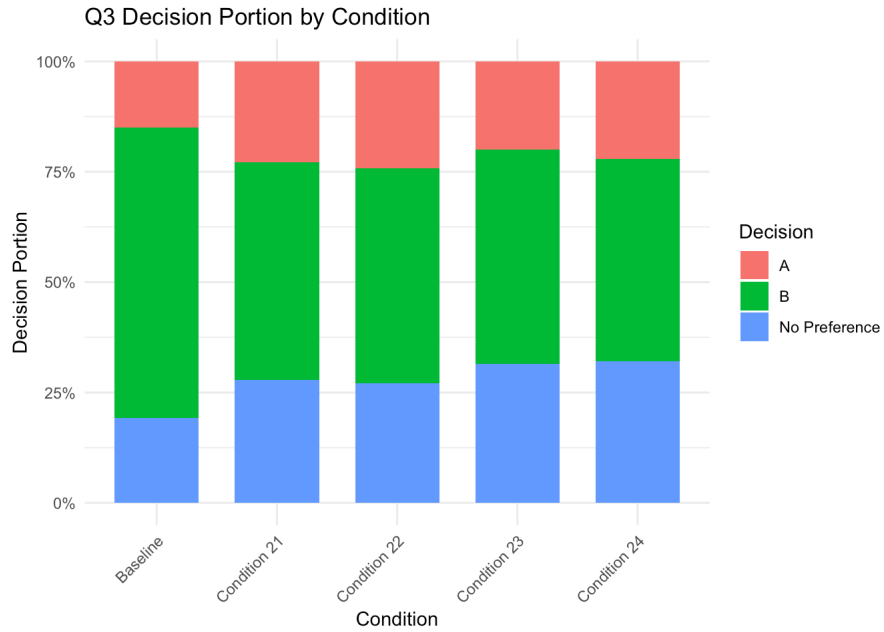
```
q123_q2_baseline_emm <- as.data.frame(
  summary(
    emmeans(
      q123_q2_lmer,
      ~ 1,
      at = list(
        is_baseline = "Yes",
        interaction_mode = "hover",
        detail_presentation = "show_one",
        vistype = "CI"
      )
    ),
    infer = c(TRUE, FALSE)
  )
) %>%
  dplyr::mutate(
    condition_label = "Baseline",
    is_baseline = "Yes"
  )

q123_q2_nonbaseline_emm <- as.data.frame(
  summary(
    emmeans(
      q123_q2_lmer,
      ~ 1,
      at = list(is_baseline = "No"),
      weights = "proportional"
    ),
    infer = c(TRUE, FALSE)
  )
) %>%
  dplyr::mutate(
    condition_label = "Non-baseline",
    is_baseline = "No"
  )

q123_q2_condition_emmeans <- dplyr::bind_rows(
  q123_q2_baseline_emm,
  q123_q2_nonbaseline_emm
) %>%
  dplyr::mutate(
    condition_label = factor(condition_label, levels = c("Baseline", "Non-baseline"))
  ) %>%
  dplyr::arrange(condition_label) %>%
  dplyr::select(condition_label, is_baseline, emmean, SE, df, lower.CL, upper.CL)

knitr::kable(
  q123_q2_condition_emmeans,
  digits = 3,
  caption = "Q2 EMMs for Baseline vs Non-baseline (Non-baseline averaged across conditions)"
)
```

Q2 EMMs for Baseline vs Non-baseline (Non-baseline averaged across conditions)

condition_label	is_baseline	emmean	SE	df	lower.CL	upper.CL
Baseline	Yes	53.053	1.124	713.147	50.846	55.261
Non-baseline	No	53.129	0.567	216.683	52.011	54.246

Q3: Decision Result (A, B, No Preference; Nominal)

```
q123_q3_model_data <- q123_q3_data %>%
  dplyr::mutate(
    decision_result = factor(decision_result, levels = c("A", "B", "No Preference")),
    interaction_availability = factor(
      dplyr::case_when(
        is_baseline == "Yes" ~ "Static",
        TRUE ~ "Interactive"
      ),
      levels = c("Static", "Interactive", "Buggy")
    ),
    vistype = factor(
      dplyr::recode(
        as.character(vistype),
        "CI" = "PI",
        "EP" = "Ensemble",
        "CIEP" = "PI + Ensemble"
      ),
      levels = c("PI", "Ensemble", "PI + Ensemble")
    )
  ) %>%
  droplevels()

condition_levels <- c(
  "Baseline",
  paste0(
    "Condition ",
    sort(unique(q123_q3_model_data$condition_number[q123_q3_model_data$is_baseline == "No"]))
  )
)

q123_q3_model_data <- q123_q3_model_data %>%
  dplyr::mutate(
    condition_label = ifelse(is_baseline == "Yes", "Baseline", paste0("Condition ", condition_number)),
    condition_label = factor(condition_label, levels = condition_levels)
  )

q3_condition_portion <- q123_q3_model_data %>%
  dplyr::count(condition_label, decision_result, .drop = FALSE, name = "n") %>%
  dplyr::group_by(condition_label) %>%
  dplyr::mutate(
    total = sum(n),
    portion = ifelse(total > 0, n / total, 0)
  ) %>%
  dplyr::ungroup()

ggplot(q3_condition_portion, aes(x = condition_label, y = portion, fill = decision_result)) +
  geom_col(position = "stack", width = 0.7) +
  scale_y_continuous(
    limits = c(0, 1),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(
    title = "Q3 Decision Portion by Condition",
    x = "Condition",
    y = "Decision Portion",
    fill = "Decision"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

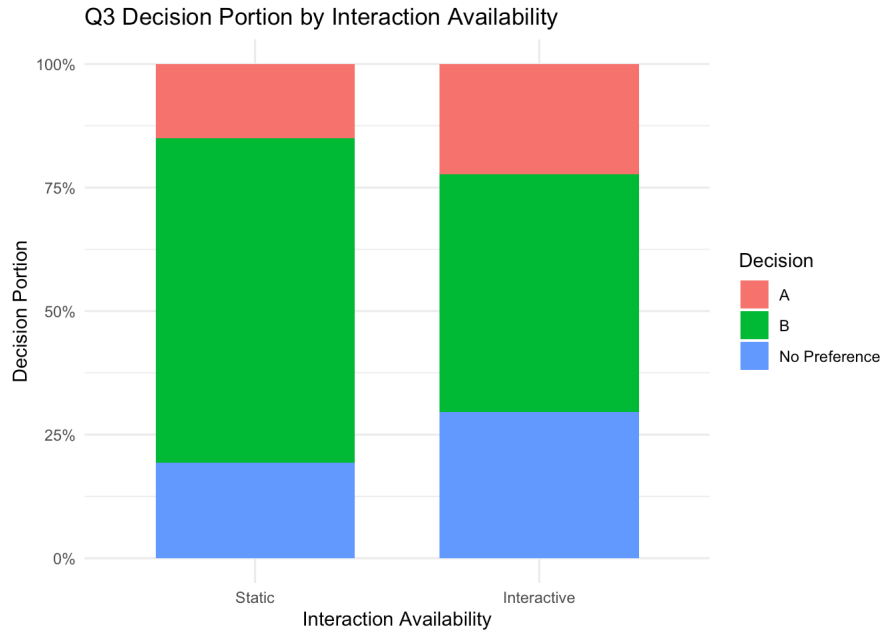



```
q3_condition_portion_table <- q3_condition_portion %>%
  dplyr::group_by(condition_label) %>%
  dplyr::summarise(
    A = portion[decision_result == "A"][1],
    B = portion[decision_result == "B"][1],
    `No preference` = portion[decision_result == "No Preference"][1],
    .groups = "drop"
  ) %>%
  dplyr::mutate(
    dplyr::across(c(A, B, `No preference`), ~ round(ifelse(is.na(.x), 0, .x), 4))
  )

knitr::kable(
  q3_condition_portion_table,
  caption = "Q3 Decision Portion Table by Condition"
)
```

Q3 Decision Portion Table by Condition

condition_label	A	B	No preference
Baseline	0.1500	0.6571	0.1929
Condition 21	0.2286	0.4929	0.2786
Condition 22	0.2429	0.4857	0.2714
Condition 23	0.2000	0.4857	0.3143
Condition 24	0.2214	0.4571	0.3214

```
q3_interaction_portion <- q123_q3_model_data %>%
  dplyr::count(interaction_availability, decision_result, .drop = FALSE, name = "n") %>%
  dplyr::group_by(interaction_availability) %>%
  dplyr::mutate(
    total = sum(n),
    portion = ifelse(total > 0, n / total, 0)
  ) %>%
  dplyr::ungroup()

ggplot(q3_interaction_portion, aes(x = interaction_availability, y = portion, fill = decision_result)) +
  geom_col(position = "stack", width = 0.7) +
  scale_y_continuous(
    limits = c(0, 1),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(
    title = "Q3 Decision Portion by Interaction Availability",
    x = "Interaction Availability",
    y = "Decision Portion",
    fill = "Decision"
  ) +
  theme_minimal()
```

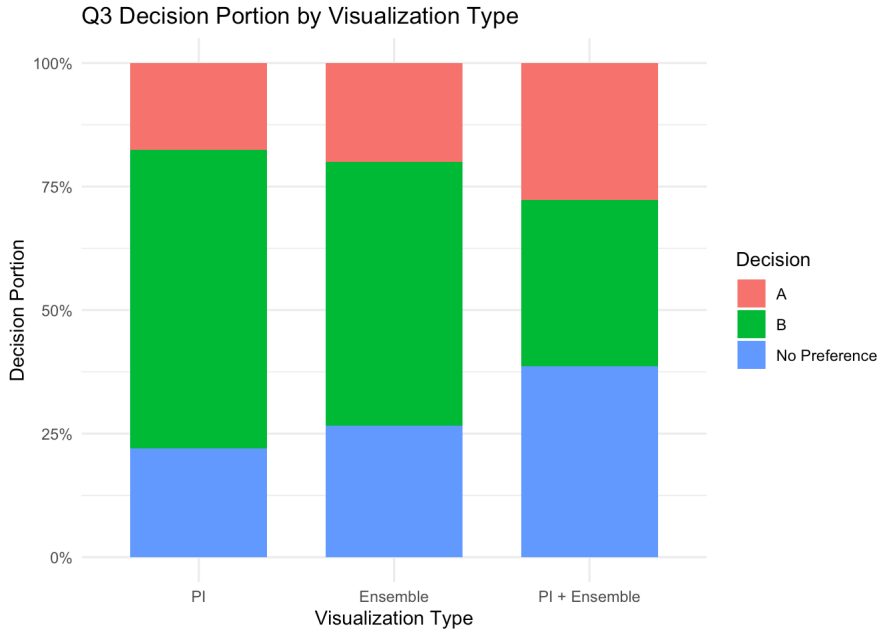



```
q3_interaction_portion_table <- q3_interaction_portion %>%
  dplyr::group_by(interaction_availability) %>%
  dplyr::summarise(
    A = portion[decision_result == "A"][1],
    B = portion[decision_result == "B"][1],
    `No preference` = portion[decision_result == "No Preference"][1],
    .groups = "drop"
  ) %>%
  dplyr::mutate(
    dplyr::across(c(A, B, `No preference`), ~ round(ifelse(is.na(.x), 0, .x), 4))
  )

knitr::kable(
  q3_interaction_portion_table,
  caption = "Q3 Decision Portion Table by Interaction Availability"
)
```

Q3 Decision Portion Table by Interaction Availability

interaction_availability	A	B	No preference
Static	0.1500	0.6571	0.1929
Interactive	0.2232	0.4804	0.2964

```
q3_vistype_portion <- q123_q3_model_data %>%
  dplyr::count(vistype, decision_result, .drop = FALSE, name = "n") %>%
  dplyr::group_by(vistype) %>%
  dplyr::mutate(
    total = sum(n),
    portion = ifelse(total > 0, n / total, 0)
  ) %>%
  dplyr::ungroup()

ggplot(q3_vistype_portion, aes(x = vistype, y = portion, fill = decision_result)) +
  geom_col(position = "stack", width = 0.7) +
  scale_y_continuous(
    limits = c(0, 1),
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  labs(
    title = "Q3 Decision Portion by Visualization Type",
    x = "Visualization Type",
    y = "Decision Portion",
    fill = "Decision"
  ) +
  theme_minimal()
```




```
q3_vistype_portion_table <- q3_vistype_portion %>%
  dplyr::group_by(vistype) %>%
  dplyr::summarise(
    A = portion[decision_result == "A"][1],
    B = portion[decision_result == "B"][1],
    `No preference` = portion[decision_result == "No Preference"][1],
    .groups = "drop"
  ) %>%
  dplyr::mutate(
    dplyr::across(c(A, B, `No preference`), ~ round(ifelse(is.na(.x), 0, .x), 4))
  )

knitr::kable(
  q3_vistype_portion_table,
  caption = "Q3 Decision Portion Table by Visualization Type"
)
```

Q3 Decision Portion Table by Visualization Type

vistype	A	B	No preference
PI	0.1756	0.6042	0.2202
Ensemble	0.2000	0.5333	0.2667
PI + Ensemble	0.2772	0.3370	0.3859